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Optimisation for decision-support

AI4NTH Summer School
13-17 July 2026

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Why does it matter?

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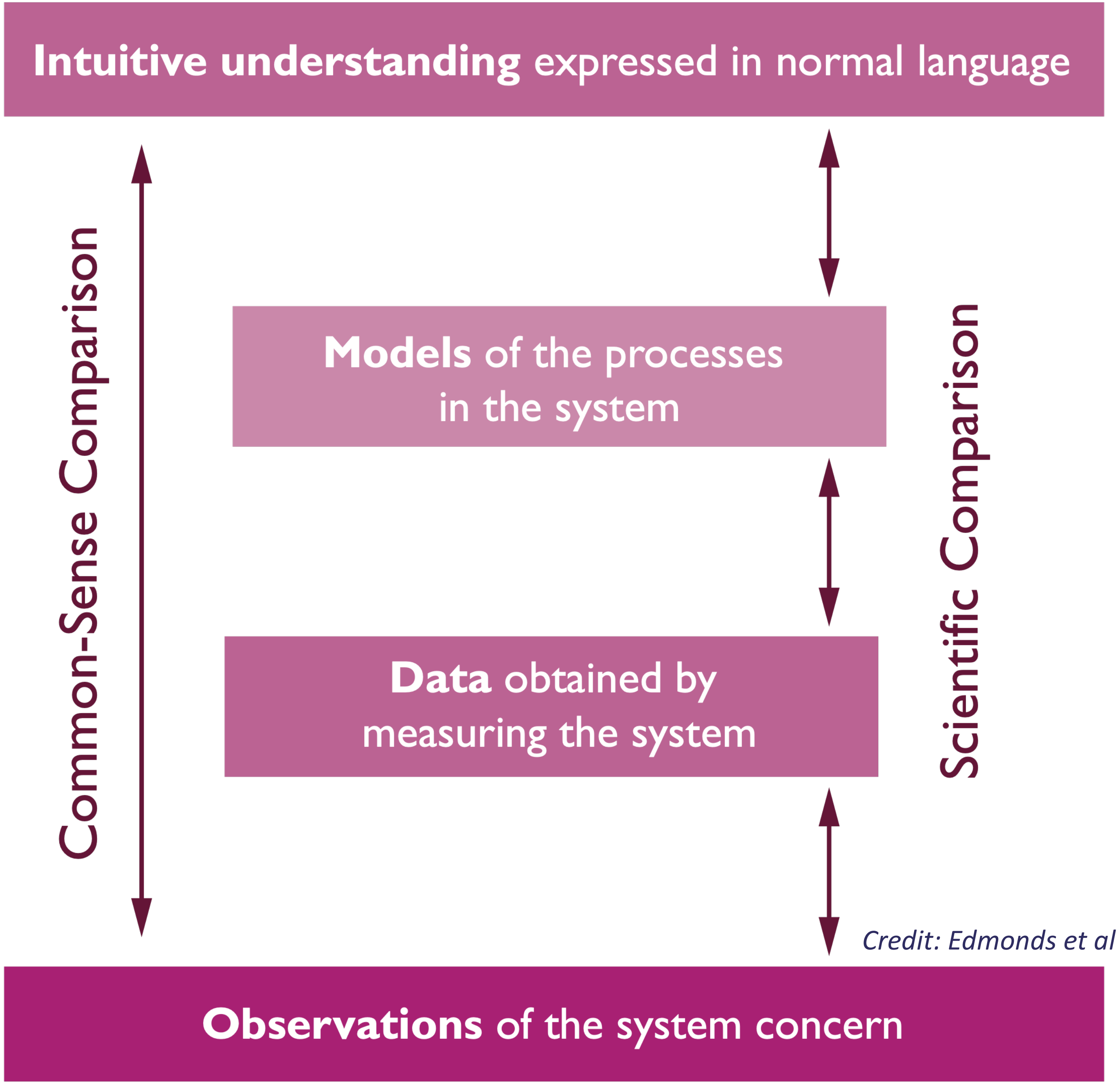
What is a model?

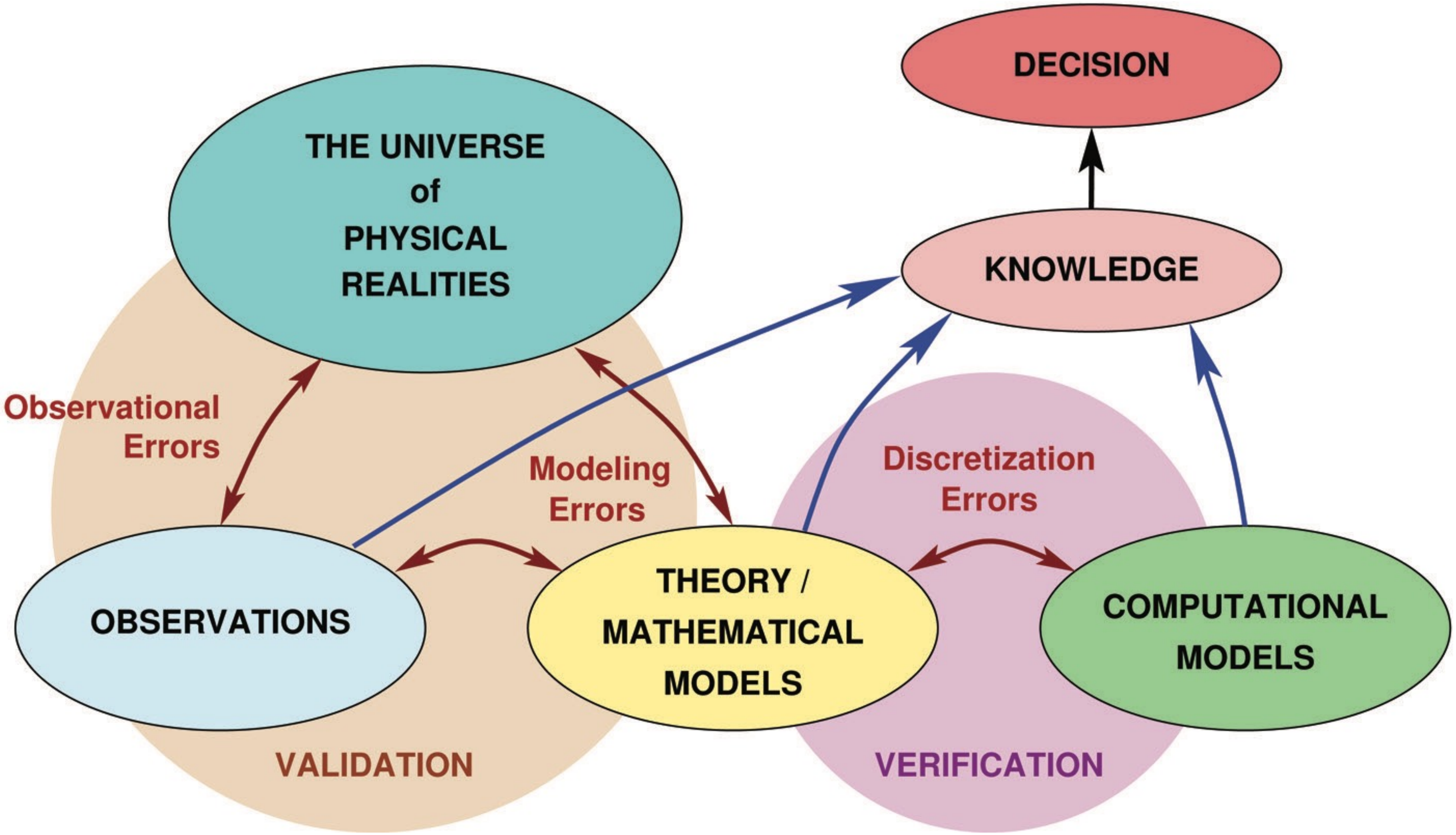
A model is simply a representation of reality.

It helps us understand a complex system.

The ultimate aim is to make better decisions.

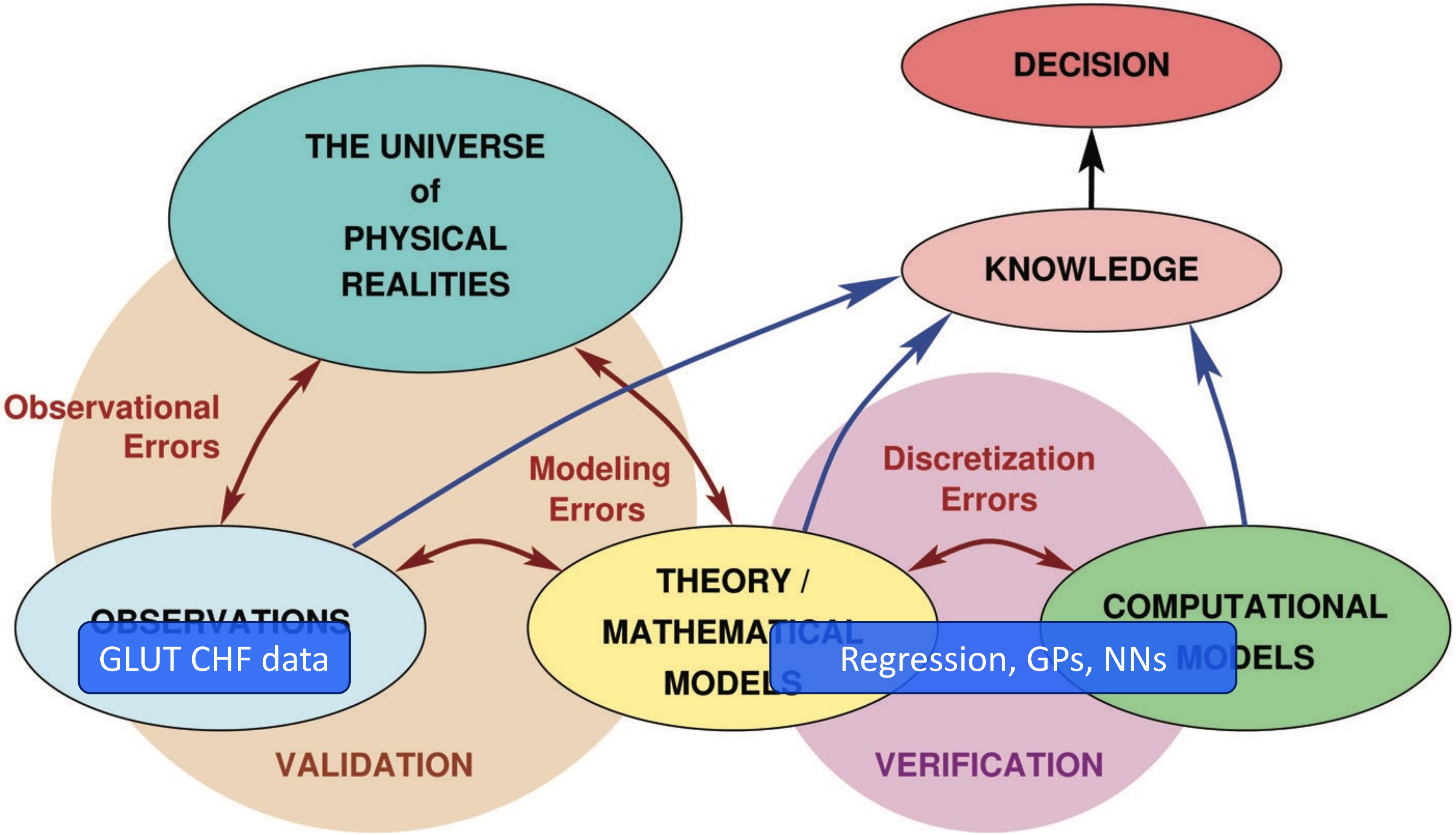


Where do models fit in?



Credit: Oden et al, SIAM News, (2010)

Where do models fit in?



Credit: Oden et al, SIAM News, (2010)

How do we use models?

"…there are known knowns; there are things we know we know. We also know there are known unknowns; that is to say we know there are some things we do not know. But there are also unknown unknowns — the ones we don't know we don't know."

- Donald Rumsfeld

	Known	Unknown
Known		
Unknown		

How do we use models?

- Awareness refers to knowledge of conditions under which model is operating
- Understanding refers to knowledge of the system embedded within the model

		Understanding	
		Known	**Unknown**
Awareness	**Known**	1 **Well-understood model and simulation**	2 **Known limitations e.g. turbulence closure**
	Unknown	3 **Unknown conditions**	4 **Rare events**

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Challenge

Given imperfect predictive models, how do we design better engineering systems?

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How do we do it?

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Breaking it down…

Given **imperfect predictive models**, how do we design better engineering systems?

- Our models tell us what an output target is for a set of input conditions.
- How does the target change with respect to changes in the inputs?

Breaking it down…

Given imperfect predictive models, how do we design **better engineering systems**?

- What does better mean?
- How do we quantify it?
- Do we have any limits/bounds to adhere to?

Into a mathematical problem

Given imperfect predictive models, how do we design **better engineering systems**?

$$\min_{x \in R^n} f(x)$$

$$s.t. g(x) \leq 0$$

$$h(x) = 0$$

$$x_L \leq x \leq x_U$$

Into a mathematical problem

Given imperfect predictive models, how do we design better engineering systems?

$$\min_{x \in R^n} f(x)$$

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Posing the problem

Given imperfect predictive models, how do we design better engineering systems?

$$\min_{x \in R^n} \boxed{f(x)}$$

$$s.t. g(x) \leq 0$$

$$h(x) = 0$$

$$x_L \leq x \leq x_U$$

Objective function

- How ‘good/better’ is our system?
- Can be a linear/nonlinear function
- Examples:
  - Pressure drop
  - Heat transfer/Mass transfer

Posing the problem

Given imperfect predictive models, how do we design better engineering systems?

$$\min_{\boxed{x \in R^n}} f(x)$$

$$s.t. g(x) \leq 0$$

$$h(x) = 0$$

$$x_L \leq x \leq x_U$$

Design variables

- What do we want to change in our system?
- Examples:
  - Shape/topology
  - Operating conditions
  - Control laws

Posing the problem

Given imperfect predictive models, how do we design better engineering systems?

$$\min_{x \in R^n} f(x)$$

$$s.t. \boxed{g(x) \leq 0}$$

$$h(x) = 0$$

$$x_L \leq x \leq x_U$$

Inequality constraints

- g(x) can be a model
- Example:
    - CHF below a certain threshold

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Posing the problem

Given imperfect predictive models, how do we design better engineering systems?

$$\min_{x \in R^n} f(x)$$

$$s.t. g(x) \leq 0$$

$$\boxed{h(x) = 0}$$

$$x_L \leq x \leq x_U$$

Equality constraints

- h(x) can be a model or physical law
- Example:
    - Navier-Stokes equation

Posing the problem

Given imperfect predictive models, how do we design better engineering systems?

$$\min_{x \in R^n} f(x)$$

$$s.t. g(x) \leq 0$$

$$h(x) = 0$$

$$\boxed{x_L \leq x \leq x_U}$$

Bounds

- Restrictions on the design variables
- Examples:
    - Limits on operating conditions
    - Limits on shape/topology

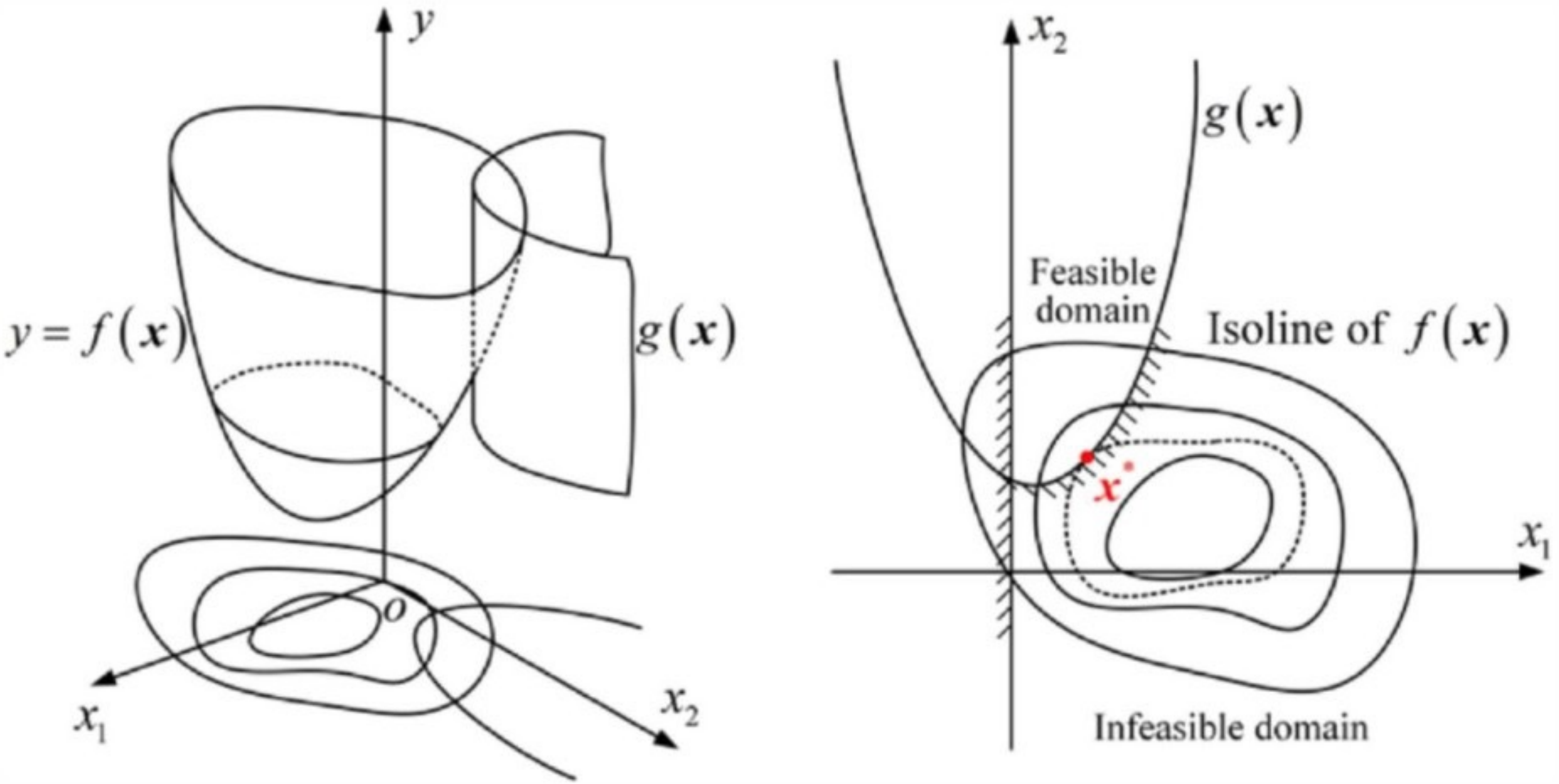
Solving the problem

$$\min_{x \in R^n} f(x)$$

$$s.t. g(x) \leq 0$$

$$h(x) = 0$$

$$x_L \leq x \leq x_U$$



Credit: https://www.linkedin.com/pulse/what-constrained-optimization-aionlinecourse-gnixf

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Solving the problem

$$\min_{x \in R^n} f(x)$$

$$s.t. g(x) \leq 0$$

$$h(x) = 0$$

$$x_L \leq x \leq x_U$$

What method should we use?

- Gradient-based methods
  - Smooth cost function
  - Differentiable models
  - Best for finding local optima
  - Examples: BFGS, SLSQP

- Gradient-free methods
  - Noisy and uncertain cost functions
  - Derivative not available or too expensive to evaluate
  - More suitable for finding global optima
  - Examples: Genetic algorithms, Bayesian Opt

Solving the problem

$$\min_{x \in R^n} f(x)$$

$$s.t. g(x) \leq 0$$

$$h(x) = 0$$

$$x_L \leq x \leq x_U$$

How do we handle the constraints?

- Lagrangian
  - Mathematically sound but needs derivatives of constraints wrt design variables
- Penalty function
  - Add component to cost function to penalize
  - Can be a hard/soft function
- Manual feasibility check

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Solving the problem

$$\min_{x \in R^n} f(x)$$

$$s.t. g(x) \leq 0$$

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How do we handle the constraints?

- Lagrangian
  - Mathematically sound but needs derivatives of constraints wrt design variables
- Penalty function
  - Add component to cost function to penalize
  - Can be a hard/soft function
- Manual feasibility check

Many modern solvers automatically handle constraints/bounds, once specified in the problem definition.

Solving the problem

$$\min_{x \in R^n} f(x)$$

$$s.t. g(x) \leq 0$$

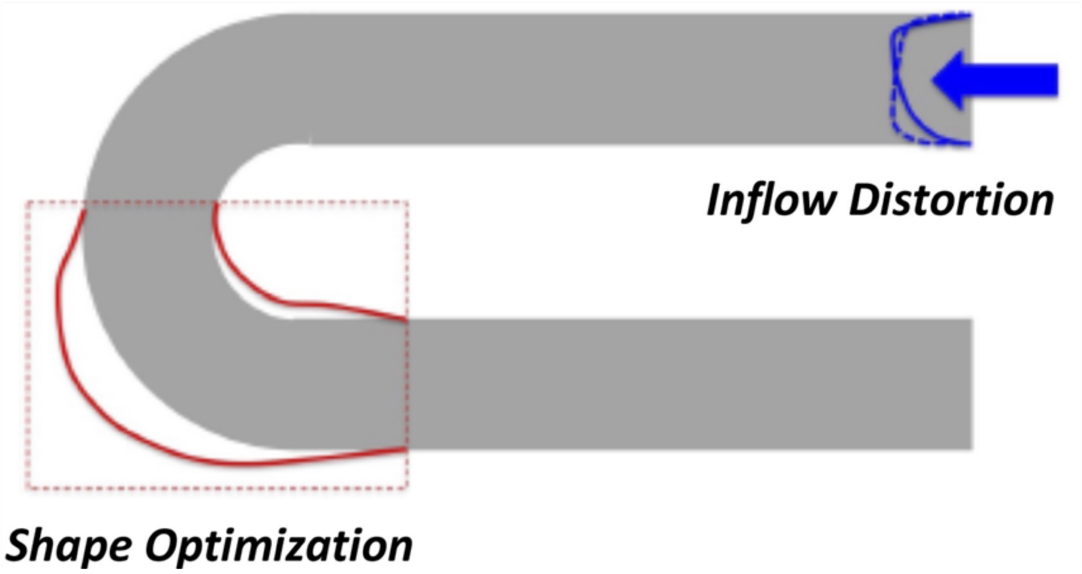
$$h(x) = 0$$

$$x_L \leq x \leq x_U$$

How do we find the next step?

- Gradient-based methods
  - Adjoint methods efficient for high number of design variables
  - ML models natively differentiable in PyTorch/JAX.
- Evolutionary methods
  - Generate candidates, keep the best, create offspring, evaluate again.
  - Progressively evolve toward better solutions.
- Bayesian method
  - Use a surrogate (GP) of the cost function to choose the next evaluation point.
  - Balance exploration of uncertain regions with exploitation of promising solutions.

Case study: Uncertainty-aware design optimisation



Design of a U-Bend

Design exploration and optimization under uncertainty

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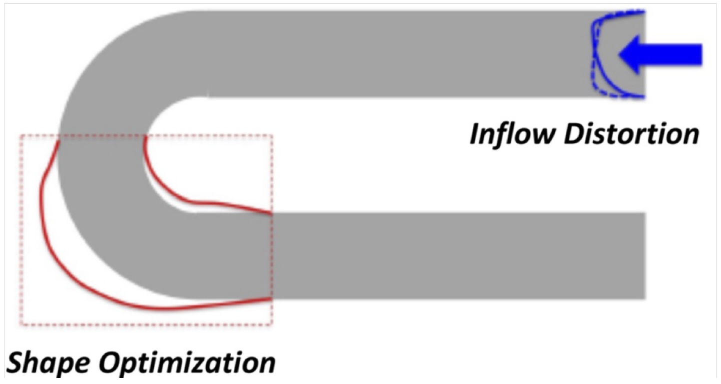
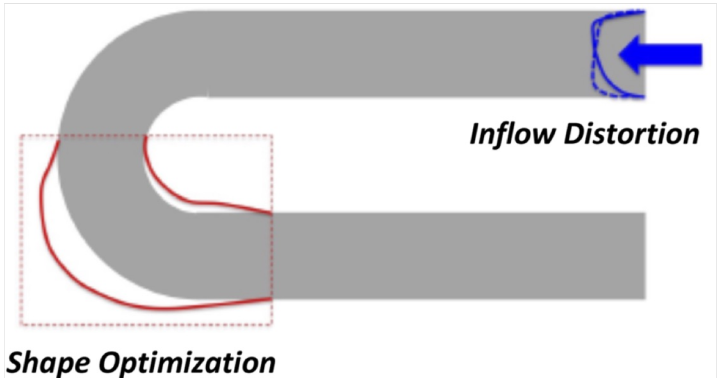
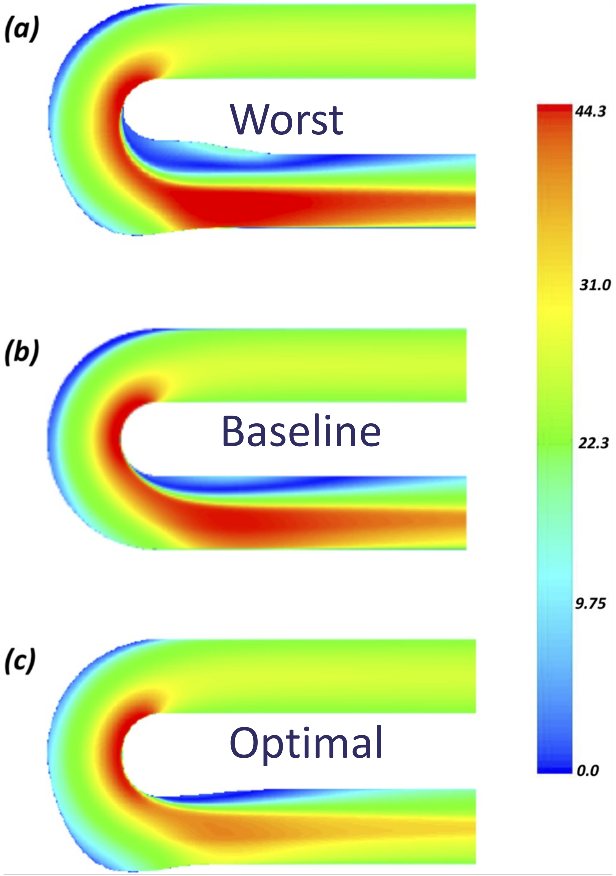
Optimisation with quantified uncertainty



- Objective: Minimize pressure loss
- Design variables: Eight control points for shape in red region
- Constraints: Turbulent Navier-Stokes equations with k-Omega SST model
- Method: Nelder-Mead algorithm
- Uncertainty: Inflow (aleatoric ~ 4%) and turbulent-model-based (epistemic ~6%)

Optimisation with quantified uncertainty

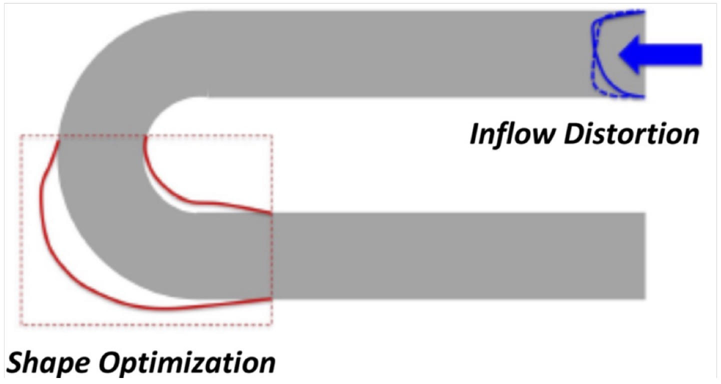




Design	Objective mean (Pa)
Worst design	191.1
Nominal design	187.3
Best design	173.8

Just optimising for the pressure:
~10% reduction in pressure loss

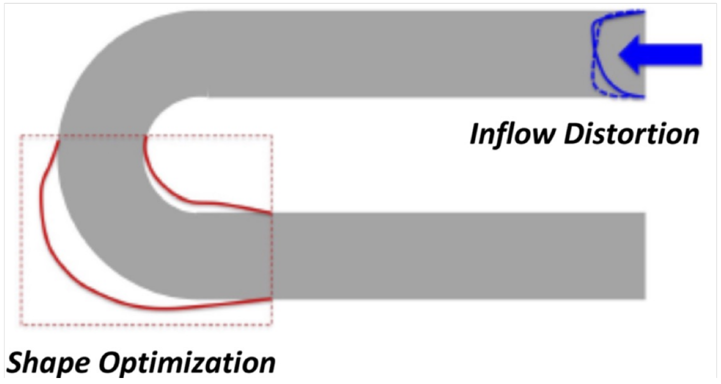
Optimisation with quantified uncertainty



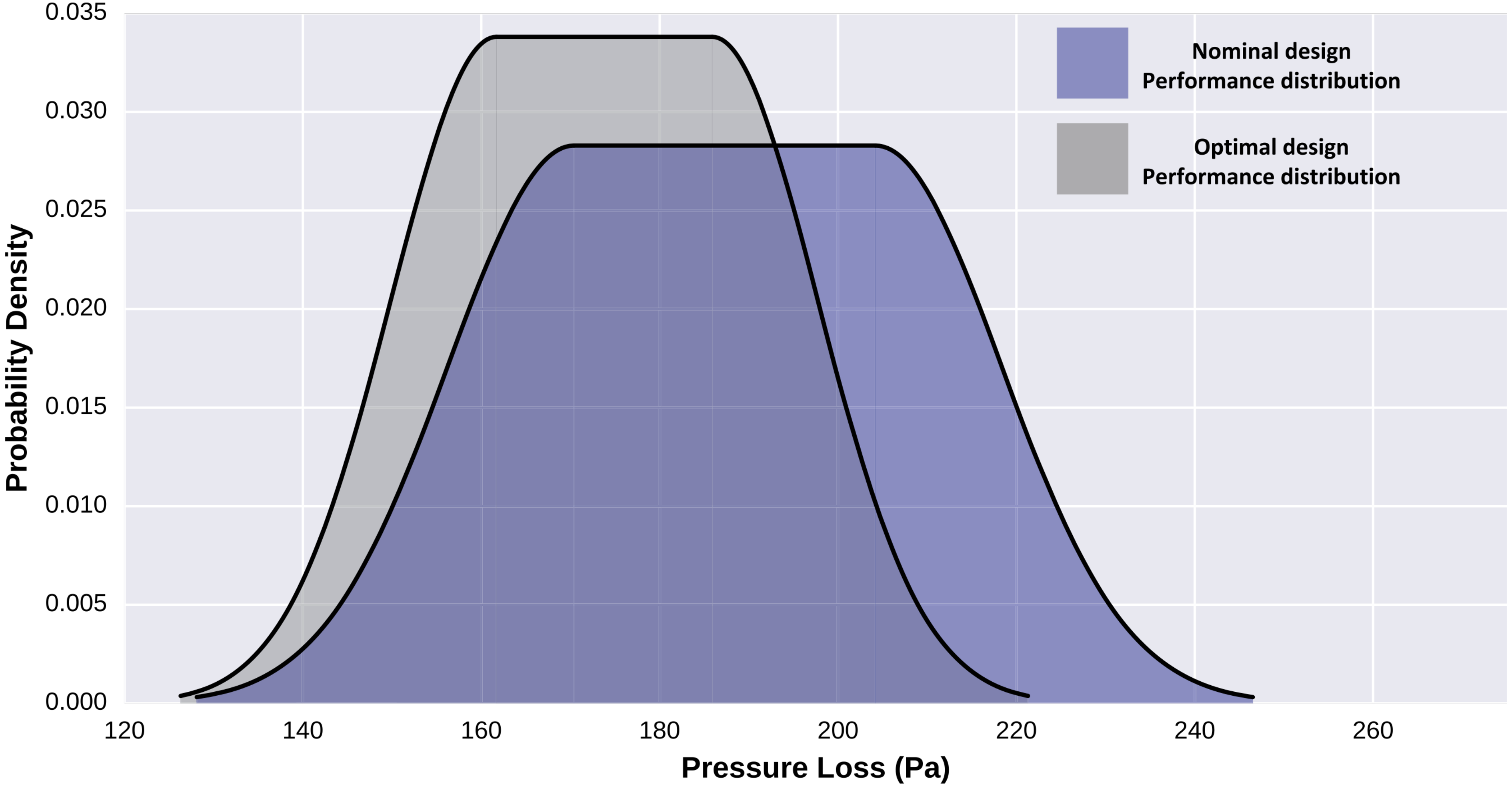
Design	Objective mean (Pa)	Epistemic variation	Aleatoric variation
Worst design	191.1	21.2	16.6
Nominal design	187.3	16.9	14.1
Best design	173.8	12.1	11.8

Accounting for the uncertainty in the pressure prediction gives us the range of values that the loss can take.

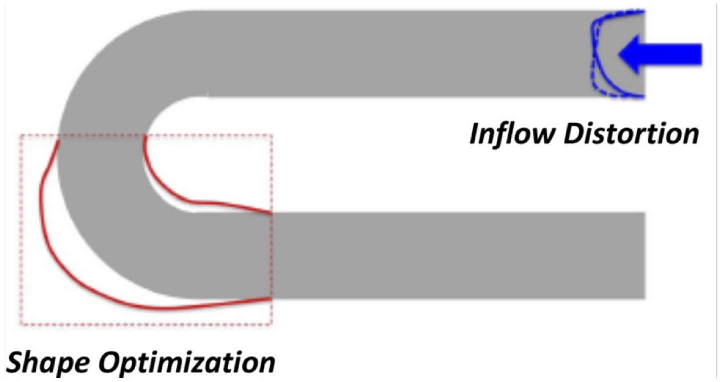
Optimisation with quantified uncertainty



The probability of benefit can guide decision making…

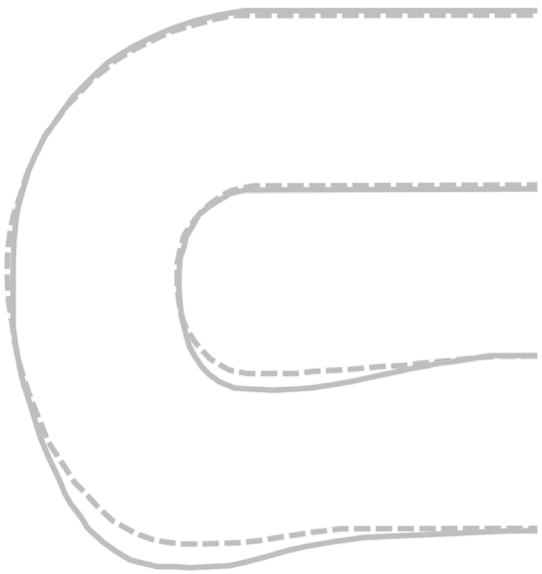


Robust optimisation under uncertainty



- Seek design that is optimal under worst-case scenarios
- Consider model-form uncertainty only
- At every step of the optimisation, consider the effect of the uncertainties in the model on the objective function and use the worst-case scenario to determine the next step

Dashed line: deterministic optimal design
Solid line: robust optimal design



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Summary

- "All models are wrong, but some are useful" – George Box. Remember the limitations of models, especially ML models.

- Optimisation requires understanding the problem, the model, and the constraints. It is a tool meant to guide decision-making not just give the solution.

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Thank you

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